

Should Siri be a Source or Medium for Ads? The Role of Source Orientation and User Motivations in User Responses to Persuasive Content from Voice Assistants

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ABSTRACT

As demonstrated by users' resistance to ads on Google Home in 2017, persuasive communications from voice assistants (VAs) can be seen as inappropriate. But, they may be better received if the VAs are not the sources but mere media for ads, as happens with radio. User motives may play a role as well, with those who use VAs primarily for information resenting ads more than those who see their VAs as social companions. To test such propositions, we conducted a scenario-based user study ($N = 264$) in which Siri acted as either a source of ads or as a medium for delivering ads by a human spokesperson. Findings suggest that for informationally motivated users, Siri as ad source causes reactance via lowered sense of control over the interaction. On the other hand, for those with social motives, it increases social presence and positively affects user experience of the interaction.

CCS CONCEPTS

• **Human-centered computing**; • **Ubiquitous and mobile devices**; • **Personal digital assistants**; • **HCI design and evaluation method**; • **Usability testing**;

KEYWORDS

Advertisement, Ad attitudes, Audio ads, Computers-are-social-actors (CASA), Psychological reactance, Smart speakers, Social presence, Virtual agents, Voice assistants

ACM Reference Format:

Eugene Cho and S. Shyam Sundar. 2022. Should Siri be a Source or Medium for Ads? The Role of Source Orientation and User Motivations in User Responses to Persuasive Content from Voice Assistants. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519667>

1 INTRODUCTION

The widespread diffusion of voice assistants (VAs) is making them promising vehicles for persuasive messages, such as advertising and health communications. However, careful consideration is needed

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CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519667>

before using VAs for advertisements. In 2017, Google faced a backlash for including a promotional message for a Disney movie in Google Home's responses [25]. No ads were broadcast after that. Amazon has been denying plans to embed ads in Alexa, while allowing ads in some types of content broadcast via Alexa, such as podcasts and streaming music services [1], which are closer to radio-type advertisements that occur during breaks in programming. It appears that when a VA serves as a medium or conduit for broadcast content, ads are tolerated, but when the VA is the source of information, e.g., when it is responding to a specific question that the user has asked, ads are considered inappropriate. In other words, contextually relevant search-engine advertising, which is the main source of revenue for companies like Google, does not appear to be feasible on VA platforms like Alexa and Siri because of user resistance to promotional content in these venues.

One reason why ads on Google Assistant were seen as inappropriate could be that users view VAs primarily as sources of information, not as media for entertainment. So, if VAs are seen more as social companions, such as social robots [23], or even audio devices like iPods [6], then perhaps ads would be more welcomed. Under this formulation, we can expect users to be more receptive to ads served by VAs, especially if the users are socially, rather than informationally, motivated to use VAs. Alternatively, users may see the delivery of ads by VA sources as a breach of trust because they are a captive audience that does not have control over the ads and may therefore resent the attempt at persuasion, especially if they are informationally, rather than socially, motivated. We explored both these possibilities with competing hypotheses, as described in the sections that follow.

1.1 VAs as Social Entities

The computers-are-social-actors (CASA) paradigm posits that users tend to treat computers as humans, based on social norms affiliated with human-to-human communication traits, even when they know they are interacting with machines [13, 18]. In the context of VAs, imbuing and strengthening basic human characteristics in virtual agents such as oral communication capabilities (e.g., even a simple feature such as human voice) can cue the "social presence heuristic" to treat digital media as a social agent than an inanimate object [20]. In support of this proposition, many users are known to personify Alexa (i.e., describe the technology as a person rather than an object), of which behaviors are linked to more social interactions and user satisfaction with Amazon Echo [17].

If the interaction with VAs can be perceived as inherently social, it is also possible that the VA delivering ad content can serve to make the interaction more personal and intimate, thus leading to

better ad attitudes. If bringing in “real people” as ad spokespeople can enhance brand perceptions by humanizing the brand [7], in alignment with CASA, we can also expect VAs to exert similar humanizing effect when they serve as ad spokespeople, compared to other formal ad mascots. As another tactic, many radio programs have their hosts endorse products (vs. radio breaks). In the same vein, VAs reading the text of an ad can effectively make them product endorsers. It is also known that media technologies such as computers and virtual agents can be considered as a distinct source to users even when the media content is not created by, or directly sourced from, the virtual beings [21].

These social responses are invariably contingent upon individual differences. For instance, Lee and Cho [10] identified users’ motives for using smart speakers, and found that the motives to engage in virtual interactions with smart speakers for escaping reality (which signaled social and personal aspects of users’ interaction with the VAs) increased positive attitudes toward the ad and the advertised brand from smart speakers. In particular, that relationship was mediated by parasocial relationships that users built with their VAs. If that is the case, people with higher social motives for using VAs could interpret VA being an ad source as more intimate, which can lead to better ad attitudes. Therefore, we predict that a VA serving as the ad endorser/source can enhance ad attitudes through heightened social presence of the interaction with the VA, but only among users with high social motives.

H1: The VA (vs. human spokesperson) delivering the ad will result in more positive ad attitudes mediated by enhanced social presence, but only among users with high social motives.

1.2 VAs as Information Sources

In contrast to the social route to positive ad attitudes, it is possible that VA acting as an ad source can result in negative ad perceptions instead. To some, native ads and personal endorsements are seen as deceptive media practices by blurring the source of advertisement [16]. Sources openly committed to persuading the audience’s view are known to trigger reactance toward persuasive messages [4], and users become more defensive and less swayable [24]. In fact, disclosure of sponsored YouTube influencer videos was found to erode advertised product desire through revealing persuasive intent, despite increasing brand memory [2]. Such negative persuasive effect of disclosure on brand attitudes was especially evident among users who have not built a strong para-social relationship with the influencer. According to Psychological Reactance Theory [3], such resistance to persuasive messages derives from feelings of threat to personal freedom and loss of control. When applied to the context of this study, since VAs are seen as information sources, their attempt to persuade users may provoke reactance because VAs may be seen as breaching the implicit contract to provide ad-free information. Moreover, by proactively delivering ads, VAs usurp control of the interaction, thus posing a threat to users’ freedom of action. This largely explains Google’s failure in 2017 [25].

But again, this tendency can differ based on users’ motives for using VAs. For instance, if the social motives are strong among users, it is likely that the negative effect of an overt persuasive intent is alleviated by the amicable relationship between users and

their VAs. Relevant to this point, even with explicit persuasive intent, likable (vs. dislikable) salespersons are found to elicit positive changes in attitudes toward the advertised product [19]. On the other hand, for users who are focused on practical use, the ad delivered by VAs can be seen as more of an interruption to the interaction than as a personal message. In fact, Lee and Cho [10] identified another major type of motive for using smart speakers, i.e., informational learning, which did not predict parasocial relationships with VAs, compared to the more social motives. For users who are informationally motivated, VA ads may appear to be hijacking their attention by pushing commercial messages into their normal information feed when they least expect them. Thus, we propose that VA serving as the ad endorser/source can induce negative ad attitudes through lowered users’ sense of control of the interaction, but only among users with high informational motives.

H2: The VA (vs. human spokesperson) delivering the ad will result in more negative ad attitudes mediated by lowered sense of control, but only among users with high informational motives.

2 METHOD

We conducted a scenario-based study with 264 participants from Amazon Mechanical Turk (MTurk) who had interacted with VAs (Amazon Alexa, Apple Siri, Google Assistant, Microsoft Cortana, Sonos One, Samsung Bixby and/or others) at least once over the past 6 months. To compare source effects, we adopted a 2 (ad source: Siri vs. ad spokesperson) X 2 (ad brand: Starbucks vs. Kleenex) mixed factorial experiment in which each participant interacted with a VA. We collected data through the end of October and early November in 2020. The sample consisted of 170 men and 94 women, age ranging from 19 to 70 ($M = 34.12$, $SD = 9.84$), with 161 Caucasians, 21 African (Americans), 64 Asian and 18 Hispanic.

2.1 Source Manipulation

We randomly assigned participants to either a condition where the ads were delivered by the VA itself ($N = 124$) or by a human ad spokesperson during the interaction with the VA ($N = 140$). We chose Siri as the VA for our study, not only because it is one of the most popular VAs in the market [14], but also because of its feature that allows it to repeat written text out loud easily to create audio clips for our stimuli. All users were asked to imagine two scenarios in random order, where they posed a question to Siri (i.e., “Siri, how do I make pumpkin spice latte?” and “Siri, how do I stop runny nose?”), each followed by responses from Siri, with contextual audio ads (for Starbucks and Kleenex respectively) embedded. For the ad stimuli, we used a 15-second audio script from the 2 brands, narrated either by a human ad spokesperson with background music or by Siri over identical background music (see Table 1 for the scripts used). In sum, both source conditions and ad topics within each source condition were randomized to control for order effects.

2.2 Measures

Manipulation Check – Ad Recall: As manipulation-check measures, users were asked to pick the kind of product and brand Siri recommended at the end of their interaction with Siri. Four options were offered for each product—Starbucks (i.e., Coffee–Dunkin,

Table 1: Stimuli scripts

Question	Main Response	VA advertisement [Siri source condition]
Siri, how do I make pumpkin spice latte?	Here's a recipe for pumpkin spice latte. Start by heating milk, pumpkin puree, sugar, pumpkin pie spices, and vanilla extract on the stove. You can use whole, 2%, dairy-free milk. Even coconut or almond milk if you feel a little adventurous. Then combine the pumpkin spice milk with strong coffee. Don't have all the ingredients to get that pumpkin spice? But, want to avoid lines in coffee shops? Instead of going through all the hassle, why don't you download the Starbucks app and stop by a store near you. Here's a promotional message from Starbucks.	We're ready to welcome you back. And we want you to know that the Starbucks app is the easiest way to find your store, order ahead, pay, and pick up your favorites. We can't wait to see you again. [Starbucks is ready to welcome you back. And Starbucks app is the easiest way to find your store, order ahead, pay, and pick up your favorites. Starbucks is waiting for you to stop by again.]
Siri, how do I stop runny nose?	Here are some home remedies to treat runny nose: Drink plenty of fluids to thin your mucus. Try using a neti pot, which is designed for nasal irrigation. Use a humidifier to further thin your mucus and add soothing moisture into the air. Are you experiencing runny nose? If so, you might also be suffering from dry and irritated skin from all the nose blowing. Why don't you try Kleenex tissues, designed to moisturize and soothe your skin? Here's a promotional message from Kleenex.	Kleenex cool touch tissues. The only tissue that releases a cool sensation to soothe a sore nose on contact. Kleenex cool touch tissues. A cool way to soothe. [Identical to original ad script]

Coffee–Starbucks, Milk–Blue Diamond Almond Breeze, Milk–Silk) and Kleenex (i.e., Cold medicine–Mucinex, Cold medicine–Tylenol, Tissue paper product–Kleenex, Tissue paper product–Charmin).

Dependent Variable – Ad Attitudes: Participants rated their agreement with 6 adjectives (i.e., good, pleasant, favorable, attractive, useful, entertaining) regarding the product recommendation offered by Siri on a 7-point scale, obtained from established scales on attitude toward the ad [11, 15] (for Starbucks: $M = 5.44$, $SD = 1.03$, $\alpha = .88$; for Kleenex: $M = 5.40$, $SD = 1.03$, $\alpha = .88$).

Mediators – Social Presence & Sense of Control: Five items from Gefen and Straub [8] were used for social presence (e.g., There was a sense of... “personalness in her (Siri's) response,” “sociability in the interaction,” “human sensitivity in her response.”) and three items from Marathe and Sundar [12] for sense of control (i.e., “Siri seemed to be responsive to the question,” “I felt in control of the interaction with Siri,” “I felt in charge of the interaction with Siri.”). All items were measured on a 7-point scale for both social presence ($M_{Starbucks} = 5.27$, $SD_{Starbucks} = 1.07$, $\alpha_{Starbucks} = .88$; $M_{Kleenex} = 5.20$, $SD_{Kleenex} = 1.12$, $\alpha_{Kleenex} = .88$) and sense of control ($M_{Starbucks} = 5.35$, $SD_{Starbucks} = 1.06$, $\alpha_{Starbucks} = .74$; $M_{Kleenex} = 5.21$, $SD_{Kleenex} = 1.11$, $\alpha_{Kleenex} = .77$).

Moderators – VA Usage Motives: Before participants listened to the audio clips, they were asked to report their social and informational motives for using VAs. For the measures, we picked seven items with highest factor loadings from each of two factors (“escaping from reality” and “informational learning”) in Lee & Cho [10] to form the social and informational motive factors respectively. As expected, two distinct factors emerged from the 14 items (see Table 2). However, we excluded 1 item that factored into a theoretically different category (i.e., “I use a smart speaker to check my schedule.”). Confirmatory factor analysis also lent support to the two-factor ($\chi^2 = 231.05$, $df = 64$, $p < .001$, CFI = .92, RMSEA = .10) over the single-factor solution ($\chi^2 = 475.38$, $df = 65$, $p < .001$, CFI = .80, RMSEA = .16) model ($\chi^2_{diff} = 244.33$, $df = 1$, $p < .001$). Due

to the high correlation ($r = .41$, $p < .001$) between social ($M = 5.45$, $SD = .87$, $\alpha = .94$) and informational ($M = 4.78$, $SD = 1.56$, $\alpha = .80$) motives, we created indices¹ that captured the relative emphasis of social (range = 13–58, $M = 45.41$, $SD = 9.96$) and informational (range = 42–87, $M = 54.59$, $SD = 9.96$) motives rather than treating them as orthogonal dimensions.

Controls – VA Usage Frequency & Brand Connection: As controls, we measured how frequently users interacted with 3 major VAs—Siri ($M = 2.11$, $SD = 1.84$), Alexa ($M = 3.85$, $SD = 2.14$) and Google Assistant ($M = 3.88$, $SD = 2.20$)—over the past 6 months, on a 7-point scale (1 = Never, 2 = Hardly ever, 3 = Occasionally, once a month, 4 = Pretty regularly, once a week, 5 = 2–3 times a week, 6 = 4–7 times a week, 7 = at least once every day). Siri usage frequency was not correlated (p 's $> .26$) with those of Alexa and GA, while the latter two were correlated ($r = .15$, $p = .02$). Brand connection to Starbucks ($M = 5.42$, $SD = 1.30$) and Kleenex ($M = 5.25$, $SD = 1.27$) were also measured on a 7-point semantic differential scale with 3 items [22] (i.e., disconnected/connected, not bonded/bonded, detached/attached).

3 RESULTS

3.1 Manipulation Check

Among the 264 participants (source assignment: 124 Siri vs. 140 ad spokesperson), 173 recalled Starbucks (83 in Siri vs. 90 in ad spokesperson conditions), and 132 recalled Kleenex (59 in Siri vs. 73 in ad spokesperson conditions) correctly. When we checked if source had any effects on ad recall (for instance, if ad details are recalled better by making the main response vs. ad content more distinct, which would be a confound), source did not affect ad recall of either Starbucks ($\text{Exp}(B) = 1.13$, $p = .65$) or Kleenex ($\text{Exp}(B) = 0.83$, $p = .46$).

¹ Social motive index = social motives ÷ (social + informational motives) × 100; Informational motive index = informational motives ÷ (social + informational motives) × 100

Table 2: Factor analyses of VA motives

Category	Items	1	2
Social Motives	I use a smart speaker to escape from reality.	.905	.039
	I use a smart speaker to forget annoying things.	.889	.080
	I use a smart speaker because I can forget complex problems.	.862	.171
	I use a smart speaker to get away from family, friends, or other issues.	.836	.089
	I use a smart speaker because I do not like feeling alone.	.828	.079
	I use a smart speaker to chat.	.817	.205
Info Motives	I use a smart speaker to engage in smart speaker-themed conversations.	.694	.406
	I use a smart speaker because I can obtain a lot of information easily.	-.203	.747
	I use a smart speaker because the information obtained using smart speakers is useful.	-.016	.732
	I use a smart speaker to obtain whatever information I need.	.206	.722
	I use a smart speaker to obtain today's news.	.331	.674
	I use a smart speaker because I can learn a lot through smart speakers	.338	.621
	I use a smart speaker to obtain information about products or services.	.364	.576
	I use a smart speaker to check my schedule.	.706	.327
Eigenvalue (Percentage of variance accounted for)		6.65 (47%)	2.32(17%)

^a Rotation method: Varimax with Kaiser Normalization.

3.2 Source Effects on Ad Attitudes through Social Presence Moderated by Social Motives

For H1, we ran moderated mediation analyses based on PROCESS Model 7 [9]. Considering the within-subjects nature of the ad context and the attendant difficulty in including a repeated-measures factor when testing moderated mediation models, we analyzed Starbucks and Kleenex ads separately. For both Starbucks and Kleenex contexts, the manipulation-check failures were filtered out from the data, leaving Starbucks = 173 and Kleenex = 132 data points. Gender, age, brand connection (Starbucks or Kleenex), and usage frequency of Siri, Alexa and GA respectively were included as controls. Social motive index was included as a moderator between source (IV) and social presence (mediator).

Starbucks Ad Context: For the Starbucks context, a significant interaction effect was found between source & social motive index on social presence ($b = .03$, $t = 2.00$, $p < .05$). Social presence was in turn significantly positively associated with attitudes toward the Starbucks ad ($b = .50$, $t = 7.63$, $p < .001$). However, the indirect effects of source on ad attitudes in the moderated mediation model as a whole was not significant (moderated mediation index = .01, 10000 bootstrap 95% CI [-.0022, .0330]). When the significant interaction was decomposed, the positive effects of social motives on social presence were more pronounced when Siri delivered the Starbucks ad compared to the human spokesperson. However, the key difference came from the Siri condition reducing social presence among users with low social motives, rather than Siri enhancing social presence (see Figure 1, left). Source alone did not have significant effects on social presence and ad attitudes (p 's $> .25$). On the other hand, social motives had a significant main effect on social presence ($b = .02$, $t = 2.84$, $p < .01$). Among controls, only Starbucks brand connection had a significant effect on ad attitudes ($b = .18$, $t = 3.18$, $p < .01$). Similar patterns appeared with analysis including those who failed the manipulation check.

Kleenex Ad Context: For the Kleenex context, the interaction between source and social motive index was marginally significant ($b = .03$, $t = 1.75$, $p = .08$), and social presence was again positively related to ad attitudes ($b = .59$, $t = 9.51$, $p < .001$). But, similar to Starbucks, the indirect source effect on Kleenex ad attitudes was not significant (moderated mediation index = .02, 10000 bootstrap 95% CI [-.0044, .0408]). The marginally significant interaction suggested that Siri (vs. ad spokesperson) increased social presence among users with high social motives as expected, but decreased social presence among those with low social motives (see Figure 1, right). Source again did not alter the levels of social presence and Kleenex ad attitudes (p 's $> .25$), while social motives had a positive main effect on social presence ($b = .04$, $t = 4.13$, $p = .001$). Kleenex brand connection was the only control that increased positive ad attitudes ($b = .12$, $t = 2.13$, $p < .05$). When the manipulation-check failures were included, the interaction between source and social motives on social presence became non-significant ($p = .12$).

3.3 Source Effects on Ad Attitudes through Sense of Control Moderated by Informational Motives

For H2, we ran the identical PROCESS Model 7 [9] with identical filters, IV (source), DV (ad attitudes) and controls as in H1, but only replaced the mediator with sense of control (vs. social presence), and the moderator with informational (vs. social) motive index. Similar patterns appeared for both ad contexts below, including manipulation check failures.

Starbucks Ad Context: This time, the indirect effect of source on ad attitudes mediated by sense of control and moderated by informational motives reached significance (moderated mediation index = -.03, 10000 bootstrap 95% CI [-.0441, -.0098]). Specifically, a significant interaction effect appeared between source and informational motives ($b = -.05$, $t = -3.52$, $p < .001$) on sense of control, which in turn was associated positively with attitudes toward the Starbucks ad ($b = .56$, $t = 9.81$, $p < .001$) (see Figure 2). As predicted,

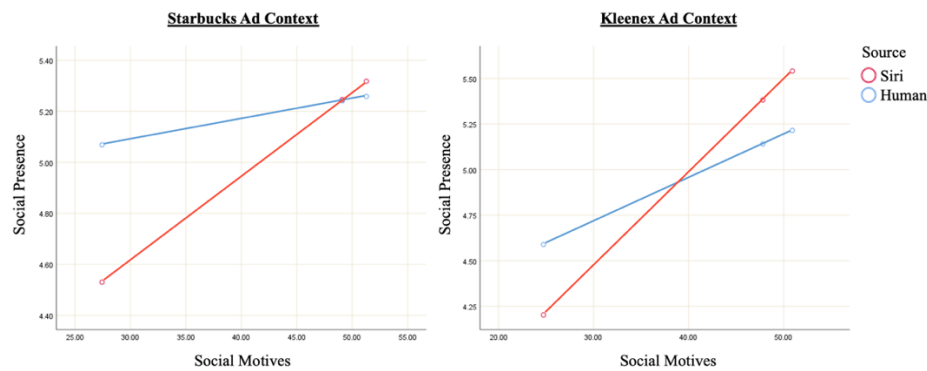


Figure 1: Interaction effect between source and social motives on social presence.

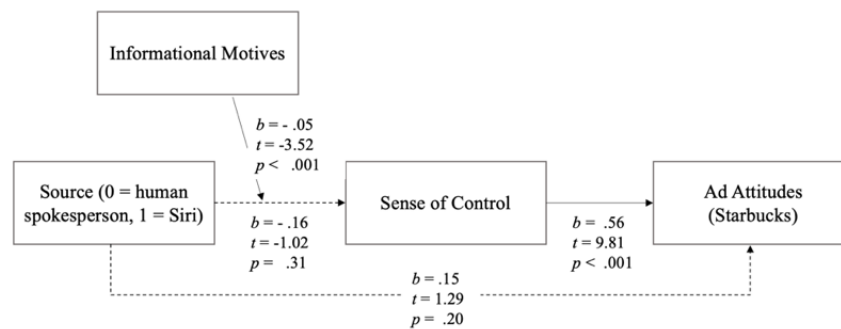


Figure 2: Indirect effects of source on ad attitudes through sense of control moderated by informational motives.

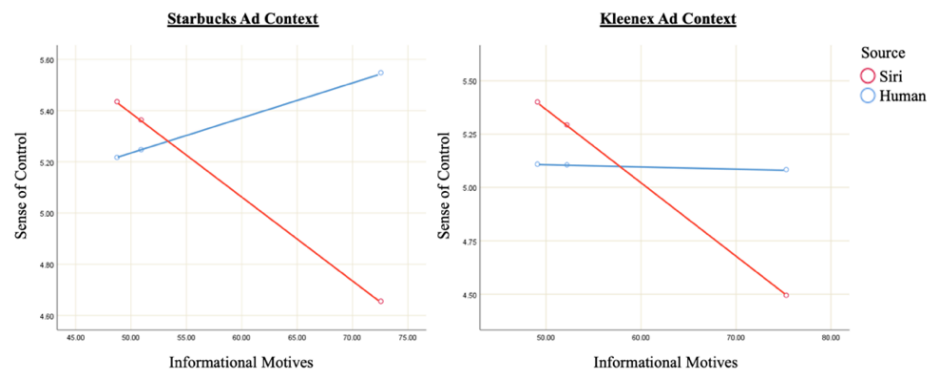


Figure 3: Interaction effect between source and info motives on sense of control for Starbucks (left) and Kleenex (right) contexts.

the significant interaction suggested that the Starbucks ad delivered by Siri drastically lowered sense of control among individuals with high informational motives (see Figure 3, left). For those low on informational motives, ad from Siri (vs. ad spokesperson) resulted in higher sense of control, but with less difference between source conditions. In sum, Starbucks ads delivered by Siri reduced sense of control among informationally motivated users, further resulting in negative attitudes toward the ad. Source did not affect sense of control ($b = -.16$, $t = -1.02$, $p = .31$) nor ad attitudes ($b = .15$, $t = 1.29$,

$p = .20$). The main effect of informational motives was also non-significant ($b = -.01$, $t = -1.14$, $p = .26$). Only brand connection with Starbucks showed a significant effect on ad attitudes as a control ($b = .21$, $t = 4.22$, $p < .001$).

Kleenex Ad Context: For the Kleenex context, the interaction between source and informational motives on sense of control was significant ($b = -.03$, $t = -2.13$, $p < .05$), and sense of control increased positive Kleenex ad attitudes ($b = .65$, $t = 10.80$, $p < .001$). However, the moderated mediation in its entirety failed to reach significance

(moderated mediation index = $-.02$, 10000 bootstrap 95% CI $[-.0438, .0017]$). The interaction pattern was similar to Starbucks, but with a flatter slope for the ad spokesperson source condition (see Figure 3, right). That is, Kleenex ads delivered by Siri (vs. ad spokesperson) lowered perceived control among informationally motivated users. The main effects of source were non-significant on sense of control ($b = -.02$, $t = -0.11$, $p = .92$), but marginally significant on Kleenex ad attitudes ($b = .22$, $t = 1.81$, $p = .07$), with informational motives showing marginally significant main effect on sense of control ($b = -.02$, $t = -1.80$, $p = .07$). Kleenex brand connection was the only control that increased ad attitudes ($b = .12$, $t = 2.28$, $p < .05$).

4 DISCUSSION

Our findings suggest that user motives and the nature of users' relationship with VAs are critical for determining the effects of persuasive communications from this new medium. As expected, we found that ads coming from Siri (vs. human spokesperson) led to negative ad attitudes via lowered sense of control, but only among those with high informational motives. This resonates with the negative effect of sponsorship disclosure on YouTube influencer-promoted brand attitudes, seen only among users disconnected with the influencer [2]. On the other hand, when Siri read out the ads, it gave rise to higher social presence (as compared to a human spokesperson) for those with high social motives. This adds nuance to Lee and Cho's [9] findings by showing that the positive effects of social motives on ad attitudes through enhanced perceived socialness can vary based on the way the ads are conveyed through VAs. While informational motives did not affect ad perceptions in their study, our results also point to the potential positive effect of informational motives on ad perceptions (if ads are delivered more effectively, targeted to the right audience).

More importantly, this suggests personalization of source orientation based on interaction history as a critical design feature that can influence user experience of VA advertisements. For informationally driven users, ads that are sourced directly from VAs can draw negative user attention and create resistance to unsolicited promotional messages. For them, VAs should remain a medium rather than a source of ad delivery, where ads are explicitly tagged as company-sponsored content. In contrast, if prior interactions have been more social in nature, users seem to be more at ease with VAs acting as ad sources without being offended by it. On a more practical note, if certain VA applications or usages are more likely to be tied to specific motives (e.g., social vs. informational), ad personalization can be easily implemented for VAs and smart speakers. For instance, when information-seeking applications are activated by users (e.g., browsing news), ads would have a better chance to be positively processed using more traditional radio-like ad breaks. Alternatively, when users engage with VAs in a more social manner featuring informal inquiries, conveying promotional messages directly through the VA may not be perceived as intrusive. Similar to how the effects of content delivery method (i.e., modality; voice vs. text) on perceived human-likeness of Microsoft's Cortana can vary by task type (i.e., utilitarian vs. hedonic) [5], the effects of ad delivery method (e.g., source orientation) on ad attitudes can be determined by specific motivation-driven usage. Perhaps Google's failure was less about the ad itself, but more about the issue of

trying to slip in a promotional message from Google Assistant pursuant to informational content such as weather, traffic, and news updates. In a similar vein, this may be why ad breaks in podcasts from Alexa are not triggering ad reactance as much. From users' perspective, such source effects in VA ads should be informed to users and carefully implemented by designers to alleviate any issues caused by manipulative advertising that is solely commercially driven without consideration of users' well-being. Considering the data collection period (Fall 2020), potential pandemic-related impact on VA usage patterns should also be recognized. To offer a more complete picture of the effectiveness of using VAs for persuasive messaging, future researchers and designers should implement and test content features (such as source and delivery style), individual differences (in use motives and relationship with VAs), as well as societal factors.

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